Design and Analysis of Randomized Trials in Public Health

Homework 4

1. Your colleagues have conducted a randomized study, but they need help analyzing the data. They have asked you to join them as a collaborator. They will pay you for your time and include you as a co-author on their publications. You agree and start working on the data. The study participants were adults recently diagnosed with a chronic disease, and the study is evaluating a new treatment compared with the standard of care. As you get to know the study better, you learn that the participants were not actually randomized to the study arms; instead, the researchers, who are physicians, examined each case and decided whether a participant should receive the new treatment or the standard of care. What do you do next? Explain your reasoning:

   a) Describe clearly what the ethical issue is here. Do you need any more information to understand the issue? If so, write a question you can ask your collaborators. (2 points)

   b) What are some possible alternative actions you can take in this situation? (4 points)

   c) Which alternative action would you choose? Which principles or frameworks would you use to make and justify your decision? (There is no single right answer here; you should explain your reasoning.) (4 points)

2. Pick one specific guideline (e.g., A1, A2, …) from the ASA's Ethical Guidelines for Statistical Practice and discuss how it relates to the conduct of randomized trials. If you like, you can use an example or paper we have used in class or draw from your own experience or reading of the literature. (4 points)

3. Please use the "hw4" SAS dataset on Brightspace to answer the questions below. Before importing the dataset into SAS, run the code below. This will create value labels for your dataset.

```
proc format;
   value yes12no     1="yes"
                     2="no";
   value sex         1="female"
                     2="male";
   value trt         1="treatment"
                     0="control";
run;
```

   These data are from a randomized trial in which eligible participants were randomized to receive a physical activity intervention ("treatment") or no intervention ("control"). Their body mass index (BMI) was measured at the beginning ("baselineBMI") and end ("endlineBMI") of the study. Participants were eligible for the study if they (1) were aged 18-50 years, (2) were evaluated to have a sedentary lifestyle, and (3) provided informed consent. Participants were excluded if they failed to meet any of these criteria.

Does the physical activity intervention affect body mass index? Use "endlineBMI" as your outcome. Follow the 10 steps of hypothesis testing we discussed in class. Are your statistical assumptions reasonable when conducting this test (present statistics or plots to support your answer)? (6 points)

Homework 4

Alexandra Chang
PUBH 526: Design and Analysis of Randomized Trials in Public Health

September 25, 2025

(Pasted from Homework 3) Your colleagues have conducted a randomized study, but they need help analyzing the data. They have asked you to join them as a collaborator. They will pay you for your time and include you as a co-author on their publications. You agree and start working on the data. The study participants were adults recently diagnosed with a chronic disease, and the study is evaluating a new treatment compared with the standard of care. As you get to know the study better, you learn that the participants were not actually randomized to the study arms; instead, the researchers, who are physicians, examined each case and decided whether a participant should receive the new treatment or the standard of care. What do you do next? Explain your reasoning:

Question 1.a (2 points)

Describe clearly what the ethical issue is here. Do you need any more information to understand the issue? If so, write a question you can ask your collaborators.

- The ethical issue is that the study was being treated and presented as a randomized controlled trial (RCT), but participants were **not actually randomized**. Instead, physicians assigned treatment based on clinical judgment. This misrepresentation raised concerns about the scientific integrity of the study and may lead to invalid causal inferences.
- The study violated the **ASA's Ethical Guidelines** below:
  - **Principle A.4**: opposes selective interpretation of data or study design
  - **Principle B.1**: requires transparency about how data were generated
  - **Principle D.5**: protects the rights and expectations of research subjects, especially if informed consent was based on randomization.
- **Yes**, I would need more information to understand the issue. I would ask the question:
  - "Did the IRB protocol and informed consent documents mention randomization, and were participants being told that their treatment would be assigned by researchers who were physicians?"

Question 1.b (4 points)

What are some possible alternative actions you can take in this situation?

- **Clarify** from the study investigators to fully understand how treatment assignment occurred and how the study was described in the IRB protocol and consent forms.
- Recommend **reclassifying** the study as observational, since treatment was assigned by physicians rather than randomly. The analysis should then use methods appropriate for observational data, such as regression adjustment and/or propensity scores.
- Advise investigators to **inform** and update the IRB about parts that have been incorrectly represented as random.
- **Withdraw** from the project if the research team refuses to correct the misrepresentation, to avoid participating in unethical scientific conduct.

Question 1.c (4 points)

Which alternative action would you choose? Which principles or frameworks would you use to make and justify your decision? (There is no single right answer here; you should explain your reasoning.)

- I would start by having a direct conversation with the research team to confirm how treatment assignments were made and whether the study was ever described as randomized in the IRB or consent forms.
  - If the participants were not actually randomized, I would recommend changing the description of the study to an observational design and adjusting the analysis accordingly.
- If the team is open to making that correction, I would be willing to stay on as a collaborator. But if they insist on presenting the study as randomized when it wasn't, I would choose to step away from the project.
  - Misrepresenting the study could lead to misleading and unethical conclusions.
- My decision reflects the ASA principles and frameworks mentioned in *Topic 3*:
  - **Recognizing the issue, weighing options, minimizing harm, and acting with transparency.**
  - Being honest about how data were collected (**Principle B**), being accountable for the way research is presented (**Principle A**), and protecting the rights of participants (**Principle D**).

Select the vignette that you found most interesting from the two webinars assigned in *Topic 3*. Then answer the following:

The following answers are based on the first vignette, **Principles A, B, C, and D.**

Question 2 (4 points)

Pick one specific guideline (e.g., A1, A2, . . .) from the ASA's Ethical Guidelines for Statistical Practice and discuss how it relates to the conduct of randomized trials. If you like, you can use an example or paper we have used in class or draw from your own experience or reading of the literature.

- **ASA Ethical Guideline B1 (Integrity of Data and Methods):**
  - Statisticians should ensure that statistical methods and procedures are clearly described and appropriate for the data and study objectives. That is, they should disclose all data collection procedures, methods of analysis, and limitations so that the work can be understood and replicated.
  - In randomized controlled trials (RCTs), the validity of the results depends heavily on the integrity of the design and analysis. ASA Guideline B1 is directly relevant as RCTs must clearly document their randomization procedures, allocation concealment, blinding, and analysis plans. If these steps are not clearly disclosed, the study's internal validity is compromised.
- **Iannotti et al. (2017):**
  - In the egg intervention trial in Ecuador discussed in Homework 3, the authors provided clear descriptions of how participants were randomized to the intervention or control group, and what statistical models (e.g., GLMs) were used to assess outcomes like stunting and growth.
  - This is a good example of their adherence to B1, which ensures transparency and reproducibility, helping readers trust the findings. Hence, ASA B1 reinforces the importance of honest, complete reporting of methods in all RCTs.
- In conclusion, without **full disclosure of methods and data integrity**, such as improperly describing non-randomized allocation as random, RCTs can mislead readers, clinicians, and policymakers.

Please use the "hw4" SAS dataset on Brightspace to answer the questions below. Before importing the dataset into SAS, run the code below. This will create value labels for your dataset.

```
proc format;
   value yes12no     1="yes"
                     2="no";
   value sex         1="female"
                     2="male";
   value trt         1="treatment"
                     0="control";
run;
```

These data are from a randomized trial in which eligible participants were randomized to receive a physical activity intervention ("treatment") or no intervention ("control"). Their body mass index (BMI) was measured at the beginning ("baselineBMI") and end ("endlineBMI") of the study. Participants were eligible for the study if they (1) were aged 18-50 years, (2) were evaluated to have a sedentary lifestyle, and (3) provided informed consent. Participants were excluded if they failed to meet any of these criteria.

Question 3 (6 points)

Does the physical activity intervention affect body mass index? Use "endlineBMI" as your outcome. Follow the 10 steps of hypothesis testing we discussed in class. Are your statistical assumptions reasonable when conducting this test (present statistics or plots to support your answer)?

- **Step 1: Scientific Question**
  - Does the physical activity intervention affect body mass index (BMI)?
- **Step 2: Define Parameters of Interest**
  - μ_1 = mean endline BMI among individuals (of our population) in the **treatment** group who receive a physical activity intervention.
  - μ_2 = mean endline BMI among individuals (of our population) in the **control** group who **did not** receive a physical activity intervention.
- **Step 3: State Null Hypothesis, H_0**
  - H_0: $\mu_1 = \mu_2$ (**no difference** in mean endline BMI between treatment and control)
    - usually an equality
  - Participants in this population have a mean endline BMI that is the **same for both** those who received the physical activity intervention and those who did not.

- **Step 4: State Directionality and Alternative Hypothesis, H_A**
  - H_A: $\mu_1 \neq \mu_2$ (there is a **difference** in mean endline BMI between groups)
    - not directional
  - Participants in this population have a mean endline BMI that is **different** between those who received the intervention and those who did not.
- **Step 5: Determine the Statistical Test and Distribution under H_0**
  - **Two-sample t-test**
    - Comparing means across two independent groups using the **t-statistic**.
    - Appropriate for outcome variables (endline BMI) that are continuous.
  - t $\mid H_0 \sim t_{n-2}$
    - Under the null hypothesis, the test statistic follows a **t-distribution**.
  - Parameters:
    - **Total Observations: 500**
    - Control Group: 186
    - Treatment Group: 217
    - Total used in analysis = 186 + 217 = **403**
      - Missing: 97
  - Degrees of Freedom:
    - **401 for pooled equal-variance**
    - **377.87 for unequal variance**
- **Step 6: State Significance Level (α), Critical Value, Define Rejection Region**
  - Significance level: α **= 0.05**
  - Critical values: $t_{\alpha/2,\ \text{df}=n-2}$, $t_{1-(\alpha/2),\ \text{df}=n-2}$
    - $t_{0.025,\ \text{df}=n-2}$, $t_{0.975,\ \text{df}=n-2}$
  - Using the equal variance method, $df \approx 401$
    - critical t-value $\approx \pm 1.96$
  - Using the unequal variance method, $df \approx 378$
    - critical t-value $\approx \pm 1.96$
  - Hence, **rejection** H_0 **if** $t < -1.96$ **or** $t > 1.96$
- **Step 7: Calculate the Test Statistic**
  - **Pooled**: t = 3.10 (equal variance)
  - **Satterthwaite**: t = 3.08 (unequal variance)
- **Step 8: Compare Test Statistic to Rejection Region, Compare p-value to α**

  - **Pooled**: t = 3.10 < 1.96, p = 0.0021 < 0.05
  - **Satterthwaite**: t = 3.08 < 1.96, p = 0.0022 < 0.05

- **Step 9: Decision on H_0**
  - **Reject** H_0 because the test statistic is in the rejection region, and the p-value is less than 0.05.

- **Step 10: Scientific Conclusion Based on Decision**
  - This study provides evidence at the 0.05 significance level that in this population, the mean endline BMI is different between participants who received the physical activity intervention and those who did not.

See SAS codes/outputs below:

```
/*************************************************************************
 * PUBH 526     Design and Analysis of Randomized Trials in Public Health
 * Homework 4
 * Alexandra Chang
 * Dataset: hw4.sas7bdat
 *************************************************************************/


/* STEP 1: Set the library */
libname mydata '/home/u64258127/sasuser.v94';

/* STEP 2: Create formats for categorical variables */
proc format;
    value yes12no 1 = "yes" 2 = "no";
    value sex     1 = "female" 2 = "male";
    value trt     1 = "treatment" 0 = "control";
run;

/* STEP 3: Check dataset structure */
proc contents data=mydata.hw4;
run;

/* STEP 4: Preview first 10 observations */
proc print data=mydata.hw4 (obs=10);
    format sex sex. treatment trt. sedentary yes12no. consent yes12no.;
    title "Table 1: First 10 Observations in hw4 Dataset";
run;

/* STEP 5: Check for missing values in treatment */
proc freq data=mydata.hw4;
    tables treatment / missing;
    format treatment trt.;
    title "Check for Missing Values in Treatment Variable";
run;

/* STEP 6: Descriptive stats of endline BMI by treatment group */
proc means data=mydata.hw4 n mean std min max;
    class treatment;
    var endlineBMI;
    format treatment trt.;
    title "Descriptive Statistics for Endline BMI by Treatment Group";
run;
```

```
42
43 /* STEP 7: Histogram and QQ plot for normality (group: control) */
44 proc univariate data=mydata.hw4 normal;
45     where treatment = 0;
46     var endlineBMI;
47     histogram endlineBMI / normal;
48     qqplot endlineBMI / normal(mu=est sigma=est);
49     title "Histogram and QQ Plot of Endline BMI (Control Group)";
50 run;
51
52 /* STEP 8: Histogram and QQ plot for normality (group: treatment) */
53 proc univariate data=mydata.hw4 normal;
54     where treatment = 1;
55     var endlineBMI;
56     histogram endlineBMI / normal;
57     qqplot endlineBMI / normal(mu=est sigma=est);
58     title "Histogram and QQ Plot of Endline BMI (Treatment Group)";
59 run;
60
61 /* STEP 9: Two-sample t-test comparing endline BMI */
62 proc ttest data=mydata.hw4;
63     class treatment;
64     var endlineBMI;
65     format treatment trt.;
66     title "Two-Sample t-test Comparing Endline BMI by Treatment
          Group";
67 run;
```

Listing 1: SAS Code for Homework 4

The CONTENTS Procedure

Data Set Name	WORK.OBS	Observations	600
Member Type	DATA	Variables	8
Engine	V9	Indexes	0
Created	01/31/2023 16:10:52	Observation Length	64
Last Modified	01/31/2023 16:10:52	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	WINDOWS_64		
Encoding	wlatin1 Western (Windows)		

Engine/Host Dependent Information	
Data Set Page Size	65536
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	1021
Obs in First Data Page	600
Number of Data Set Repairs	0
ExtendObsCounter	V9
Filename	/home/u60439012/Dataset.sfd/had.sas7bdat
Release Created	9.0401M7
Host Created	X64_SUPRO
Inode Number	1079153288
Access Permission	rw-r--r--
Owner Name	sas.u60439012.u1
File Size	128KB
File Size (bytes)	131072

Alphabetic List of Variables and Attributes

#	Variable	Type	Len	Format	Label
1	id	Num	8		Participant ID
2	bakersid000	Num	8		Bakersfield Risk
3	consent	Num	8	YESNO.	Provided Informed Consent
4	endobmi000	Num	8		Endline BMI
5	sc	Num	8		Participant SI
6	sedentary	Num	8	YESNO.	Sedentary Lifestyle
7	sex	Num	8	SEX.	Participant Sex
8	treatment	Num	8	TRT.	Treatment

Table 1: First 10 Observations in had Dataset

Obs	id	age	sedentary	consent	sex	treatment	baselinebmi	endlinebmi
1	1	40	yes	yes	male	treatment	27.1	26.2
2	2	41	yes	yes	male	treatment	28.6	27.2
3	3	26	yes	yes	female	control	27.9	25.2
4	4	28	yes	yes	male	treatment	28.4	24.3
5	5	28	yes	yes	female	treatment	25.4	23.2
6	6	35	yes	yes	female	control	28.9	25.6
7	7	33	yes	yes	male	treatment	27.8	27.1
8	8	35	yes	yes	male	control	29.5	28.1
9	9	40	yes	yes	female	treatment	30.1	28.7
10	10	33	yes	yes	male	treatment	25.6	23.8

Check for Missing Values in Treatment Variable

The FREQ Procedure

Treatment				
treatment	Frequency	Percent	Cumulative Frequency	Cumulative Percent
	87	14.5	87	14.5
control	199	37.23	283	56.83
treatment	217	43.42	500	100.00

Descriptive Statistics for Endline BMI by Treatment Group

The MEANS Procedure

Analysis Variable : endlinebmi Endline BMI

Treatment	N Obs	N	Mean	Std Dev	Minimum	Maximum
control	199	199	26.6299899	2.0577198	21.6000000	32.9000000
treatment	217	217	25.8238431	1.9977021	21.1000000	32.4000000

Histogram and QQ Plot of Endline BMI (Control Group)

The UNIVARIATE Procedure

Variable: endlinebmi (Endline BMI)

Moments			
N	199	Sum Weights	199
Mean	26.6299899	Sum Observations	4993.1
Std Deviation	2.05771198	Variance	4.23455533
Skewness	0.31691509	Kurtosis	0.09957628
Uncorrected SS	134000.75	Corrected SS	775.79168
Coeff Variation	7.72654567	Std Error Mean	0.15404304

Basic Statistical Measures			
Location		Variability	
Mean	26.62999	Std Deviation	2.05771
Median	26.60000	Variance	4.23455
Mode	26.70000	Range	11.30000
		Interquartile Range	2.80000

Tests for Location: Mu0=0			
Test	Statistic		p Value
Student's t	t	177.2121	Pr > |t| <.0001
Sign	M	99.5	Pr >= |M| <.0001
Signed Rank	S	9950.5	Pr >= |S| <.0001

Tests for Normality			
Test	Statistic		p Value
Shapiro-Wilk	W	0.991114	Pr < W 0.3275
Kolmogorov-Smirnov	D	0.042373	Pr > D >0.1500
Cramer-von Mises	W-Sq	0.042372	Pr > W-Sq >0.2500
Anderson-Darling	A-Sq	0.370384	Pr > A-Sq >0.2500

Quantiles (Definition 5)	
Level	Quantile
100% Max	32.9
99%	32.5
95%	31.3
90%	29.4
75% Q3	27.8
50% Median	26.6
25% Q1	25.1
10%	23.7
5%	23.0
1%	21.7
0% Min	21.6

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
21.6	165	30.8	194
22.1	101	30.7	78
22.8	246	32.3	313
22.9	249	32.4	462
23.0	227	32.9	398

Histogram and QQ Plot of Endline BMI (Control Group)

The UNIVARIATE Procedure

Fitted Normal Distribution for endlinebmi (Endline BMI)

Parameters for Normal Distribution		
Parameter	Symbol	Estimate
Mean	Mu	26.62997
Std Dev	Sigma	2.05772

Goodness-of-Fit Tests for Normal Distribution			
Test	Statistic		p Value
Kolmogorov-Smirnov	D	0.042373	Pr > D >0.150
Cramer-von Mises	W-Sq	0.042372	Pr > W-Sq >0.250
Anderson-Darling	A-Sq	0.370384	Pr > A-Sq >0.250

Quantiles for Normal Distribution		
	Quantile	
Percent	Observed	Estimated
1.0	22.100	21.838
5.0	23.600	23.248
10.0	24.100	24.002
25.0	25.100	25.247
50.0	26.600	26.630
75.0	27.800	28.013
90.0	29.400	29.256
95.0	31.300	30.043
99.0	32.400	31.428

Histogram and QQ Plot of Endline BMI (Control Group)

The UNIVARIATE Procedure

Q-Q Plot for endlinebmi

Histogram and QQ Plot of Endline BMI (Treatment Group)

The UNIVARIATE Procedure

Variable: endlinebmi (Endline BMI)

Moments			
N	217	Sum Weights	217
Mean	25.8238431	Sum Observations	5604.2
Std Deviation	1.99772712	Variance	3.99094459
Skewness	-0.1005582	Kurtosis	-0.256639
Uncorrected SS	147716.52	Corrected SS	863.49036
Coeff Variation	7.73654114	Std Error Mean	0.13559454

Basic Statistical Measures			
Location		Variability	
Mean	25.82384	Std Deviation	1.99771
Median	25.80000	Variance	3.99094
Mode	24.50000	Range	11.30000
		Interquartile Range	2.60000

Note: The mode displayed is the smallest of 2 modes with a count of 5.

Tests for Location: Mu0=0			
Test	Statistic		p Value
Student's t	t	190.4154	Pr > |t| <.0001
Sign	M	108.5	Pr >= |M| <.0001
Signed Rank	S	11826.5	Pr >= |S| <.0001

Tests for Normality			
Test	Statistic		p Value
Shapiro-Wilk	W	0.994071	Pr < W 0.6105
Kolmogorov-Smirnov	D	0.034911	Pr > D >0.150
Cramer-von Mises	W-Sq	0.027158	Pr > W-Sq >0.250
Anderson-Darling	A-Sq	0.21496	Pr > A-Sq >0.250

Quantiles (Definition 5)	
Level	Quantile
100% Max	30.4
99%	29.1
95%	29.1
90%	28.5
75% Q3	27.4
50% Median	26.0
25% Q1	24.8
10%	23.5
5%	22.9
1%	21.7
0% Min	21.1

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
21.1	223	30.1	378
21.4	335	30.2	51
21.7	333	30.9	13
21.8	177	30.2	276
22.2	437	30.4	449

Histogram and QQ Plot of Endline BMI (Treatment Group)

The UNIVARIATE Procedure

Distribution of endlinebmi

Histogram and QQ Plot of Endline BMI (Treatment Group)

The UNIVARIATE Procedure

Fitted Normal Distribution for endlinebmi (Endline BMI)

Parameters for Normal Distribution		
Parameter	Symbol	Estimate
Mean	Mu	25.82384
Std Dev	Sigma	1.997727

Goodness-of-Fit Tests for Normal Distribution			
Test	Statistic		p Value
Kolmogorov-Smirnov	D	0.0349091	Pr > D >0.150
Cramer-von Mises	W-Sq	0.027158	Pr > W-Sq >0.250
Anderson-Darling	A-Sq	0.214962	Pr > A-Sq >0.250

Quantiles for Normal Distribution		
	Quantile	
Percent	Observed	Estimated
1.0	21.100	21.016
5.0	22.600	22.919
10.0	23.500	23.634
25.0	24.800	24.764
50.0	26.000	25.845
75.0	27.400	27.207
90.0	28.500	28.010
95.0	29.100	28.483
99.0	29.800	30.389

Histogram and QQ Plot of Endline BMI (Treatment Group)

The UNIVARIATE Procedure

Q-Q Plot for endlinebmi

Two-Sample t-test Comparing Endline BMI by Treatment Group

The TTEST Procedure

Variable: endlinebmi (Endline BMI)

treatment	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
control		199	26.6299	2.0577	0.1459	21.6000	32.9000
treatment		217	25.8240	1.9977	0.1356	21.1000	30.4000
Diff (1-2)	Pooled		0.8059	1.9994	0.1903		
Diff (1-2)	Satterthwaite		0.8059		0.1897		

treatment	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
control		26.6299	26.3420	26.9200	2.0577	1.8623	2.2944
treatment		25.8240	25.7761	26.2719	1.9977	1.8070	2.2522
Diff (1-2)	Pooled	0.8059	0.2219	0.9899	1.9948	1.8283	2.1931
Diff (1-2)	Satterthwaite	0.8059	0.2199	0.9920			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	407	3.70	0.0002
Satterthwaite	Unequal	377.97	3.98	0.0002

Equality of Variances

Method	Num DF	Den DF	F Value	Pr > F
Folded F	198	216	1.06	0.5405

Distribution of endlinebmi

Q-Q Plots of endlinebmi